

Mohammed Abdu

Full Stack Engineer

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Results-driven Lead Software Engineer with 8+ years of expertise in architecting and delivering scalable web applications, microservices, and enterprise-grade backend systems. Proven track record of leading cross-functional teams, optimizing system performance by up to 68%, and executing robust solutions that serve millions of users. Expert in Java ecosystem, Spring Boot, cloud technologies, and modern development practices with demonstrated success in fintech, e-commerce, and government sectors.

Education

Bachelor of Engineering in Systems and Computer Engineering at Al-Azhar University, Cairo, Egypt

October 2012 - May 2017

Experience

Lead Full Stack Engineer at Dubai Police HQ, Dubai, UAE

July 2023 - Present • [↗](#)

- Led development of mission-critical web applications ensuring 99.9% uptime.
- Architected the Traffic System with real-time data processing and operational dashboards.
- Executed a Spring Boot architecture reducing downtime by 25% and improving performance by 30%.
- Enhanced frontend with Vue.js and Vuetify 3, increasing user engagement by 40%.
- Integrated WebSocket communication features boosting user interaction by 20%.
- Implemented Arabic/English i18n support, expanding user base by 40%.
- Migrated legacy Spring MVC apps to Spring Boot for boosted scalability.

Software Engineer at IKEASA, El Maadi, Cairo, Egypt

March 2022 - July 2023 • [↗](#)

- Engineered centralized logging and optimized performance via AKAMAI CDN for 1M+ users.
- Revamped CMS platform, improving content creation efficiency by 30%.
- Automated backend processes using Azure Functions, reducing operational overhead.
- Built an error alert system cutting resolution time by 40%.
- Resolved critical production issues within 24-hour SLA.

Lead Full Stack Engineer at BOUTIQAAT, Cairo, Egypt (Part-time)

January 2022 - May 2023 • [↗](#)

- Enhanced the enterprise admin portal with PCI-compliant payment integrations, resulting in a 35% reduction in transaction errors and enabling secure processing of over 10,000 monthly payments.
- Spearheaded coupon system increasing customer engagement by 25%.
- Overhauled Catalog Admin infrastructure, reducing deployment time by 80%.
- Restructured core modules, achieving 68% performance gains.

Senior Microservices Engineer at Mashreq Bank, Dubai, United Arab Emirates (Remote)

July 2021 - December 2021 • [↗](#)

- Streamlined banking system operations for mobile app performance.
- Upgraded legacy software, improving code efficiency by 15%.
- Deployed Splunk for centralized log analytics, accelerating issue resolution by 30%.
- Pioneered test automation, reducing repetitive tasks by 40%.

Java Microservices Engineer at IDEALWAVES, Cairo, Egypt

May 2019 - June 2021 • [↗](#)

- Developed scalable e-commerce platform handling millions of transactions.
- Created patent-pending innovation increasing conversion rates by 20%.
- Restructured cloud infrastructure, saving 30% in costs.
- Engineered event-driven notification system with Apache Camel.
- Mentored junior developers on microservices architecture.

Software Engineer at Freelance, Remote

January 2017 - December 2019

- Delivered 3+ enterprise solutions for education and healthcare sectors serving +10,000 daily users.
- Reduced manual processes by 40% through web applications.

Embedded Systems Software Engineer at EmbeddedFab, Al Jizah, Egypt

July 2016 - December 2017 • [↗](#)

- Developed embedded C/C++ solutions for IoT devices improving processing speed by 30%.

- Integrated sensor networks reducing data latency by 60%.
- Improved firmware power consumption by 40%.
- Led PCB design reviews reducing costs by 25%.

Skills

Programming Languages: Java SE (8, 11, 17, 21, 24), Kotlin, JavaScript, TypeScript, Node.js, C, Embedded C, Golang, C#

Frameworks & Platforms: Spring Boot (2 & 3 & 4), Spring Cloud, Spring Security, Spring Batch, Spring Data, Spring Session, Hibernate, JPA, Apache Spark, React.js, Vue.js

Cloud & DevOps: AWS, Azure, DigitalOcean, Docker, Kubernetes, Portainer, CI/CD, Azure DevOps, Git, Maven

Microservices & Integration: REST APIs, SOAP, WebSocket, Apache Kafka, Apache Camel, OAuth2, JWT, Keycloak, Microservices Architecture, Domain-Driven Design (DDD)

Databases & Persistence: MySQL, PostgreSQL, Oracle, SQL Server, MongoDB, Azure Cosmos DB, Redis, Liquibase, Flyway

Monitoring, Logging & Testing: Splunk, Prometheus, Grafana, Elasticsearch, Azure Monitor, SonarQube, JUnit 5, Postman, JMeter, Test-Driven Development (TDD)

Development Tools & IDEs: IntelliJ IDEA, WebStorm, PyCharm, Cursor, VS Code, Eclipse, Spring Tool Suite (STS), FileZilla, PuTTY, WinSCP, VirtualBox

Application Servers & Runtimes: WebLogic, Tomcat

Project & Collaboration Tools: Jira, Confluence, ZOHO Sprints, Trello, Asana, Monday.com, Slack, Microsoft Teams, Zoom, Notion, Google Workspace (Docs, Sheets, Meet), WIKI

Operating Systems: macOS (Primary), Linux, Windows

Volunteering

Technical Mentor at MSP Tech Club - Al Azhar University, Cairo, Egypt

January 2013 - July 2017

- Guided and coached 10+ junior developers in modern web development and system architecture, improving their coding skills and accelerating project delivery by 30%.

Projects

Book Store • [↗](#)

Spring Boot 3 backend with RESTful APIs for simple book store.

Vue-Todo • [↗](#)

To-Do List application enabling users to create, edit, and delete tasks efficiently, resulting in a 25% increase in task management productivity for users.

SOAP Web Service Demo • [↗](#)

Java SOAP web service with Spring Boot and JAX-WS.

SOAP Client Demo • [↗](#)

Java SOAP client consuming external web services with secure communication.

Languages

Arabic - Native

English - Proficient

Spanish - Basic

Certificates

Oracle Certified Professional: Java SE 11 Developer from Oracle

April 2021 • [↗](#)

Demonstrated advanced proficiency in Java SE 11, covering OOP, modularity, functional programming, exception handling, multithreading, and performance optimization.